DES-notes - Calculus cheat sheet

Differential Equations (University of Melbourne)



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Trial solutions for constant coefficient ODEs (revision)

$$\mathsf{T} = \frac{\mathrm{d}^2}{\mathrm{d}t^2} + b\frac{\mathrm{d}}{\mathrm{d}t} + c, \qquad \text{where } b, c \in \mathbb{R}$$

Recall:

- Solving $\mathsf{T}y = 0$ begins with trialling a solution of the form $y(t) = \mathrm{e}^{\lambda t}$, where $\lambda \in \mathbb{C}$.
- This leads to the characteristic equation $\lambda^2 + b\lambda + c = 0$
- Then $\lambda_{\pm 2} = \dfrac{-b \pm \sqrt{b^2 - 4c}}{2}$
- Therefore, if $b^2 - 4c \neq 0$, $y = C_1\mathrm{e}^{\lambda_1 t} + C_2\mathrm{e}^{\lambda_2 t}$

Second order linear ODEs

General form $(\mathsf{T}y)(t) = g(t)$ where $\mathsf{T} = \dfrac{\mathrm{d}^2}{\mathrm{d}t^2} + f_1(t)\dfrac{\mathrm{d}}{\mathrm{d}t} + f_0(t)$.

- No general method to find solutions.
- We will use trial solutions, reduction of order, matrix diagonalisation (systems of ODEs) and (in Topic 2) power series to find solutions.
- Linear algebra analogy and solution structure holds.
- Dimension of the kernel of T is 2.

The same procedure works for linear ODEs.

- Solve $\mathsf{T}y_c = 0$ (complementary solutions are the kernel of T).
- Find a single solution to $\mathsf{T}y_p = g$.
- General solution: $y = y_c + y_p$.

Recall exercise 1.3 on slide 11

- complementary solution: $y_c = Ct$.
  One unknown constant so the dimension of the kernel of T is one.
- Particular solution: $y_p = t\log t$.
- General solution: $y = Ct + t\log t$

$$\boldsymbol{\mathcal{I}(x) = e^{\int \mathcal{P} dx}}$$

A linear first order ODE has the form:

$$\frac{dy}{dx} + \mathcal{P}(x)y = \mathcal{Q}(x)$$

To solve:

Multiply ODE by $\mathcal{I}(x)$

$$\mathcal{I}(x)\frac{dy}{dx} + \mathcal{P}(x)\mathcal{I}(x)y = \mathcal{Q}(x)\mathcal{I}(x)$$

If the left side can be written as the derivative of $y(x)\mathcal{I}(x)$, then

$$\frac{d}{dx}\left[y(x)\mathcal{I}(x)\right] = \mathcal{Q}(x)\mathcal{I}(x)$$

which can be solved by integrating with respect to x.

Integration by Parts

The product rule for differentiation is

$$\frac{d}{dx}(uv) = \frac{du}{dx}v + u\frac{dv}{dx}$$

Integrate

$$\int \frac{d}{dx}(uv)\,dx = \int \left(\frac{du}{dx}v + u\frac{dv}{dx}\right)dx$$

$$\Rightarrow uv = \int \frac{du}{dx}v\,dx + \int u\frac{dv}{dx}\,dx$$

$$\Rightarrow \boxed{\int u\frac{dv}{dx}dx = uv - \int v\frac{du}{dx}dx}$$

1. Integrating factor (revision)
2. Trial solutions
3. Reduction of order
4. Diagonalisation

Definition

1. Solutions to ODEs contain unknown constants.
2. Constants can be fixed by specifying *initial values* at a single value, t_0, of the solution y and its derivatives.
3. ODEs of order n require n initial values.

Definition

A *linear ODE* is a linear combination of derivatives of an unknown function y whose coefficients may be functions of t but not y:

$$f_n(t)y^{(n)}(t) + \ldots + f_1(t)y'(t) + f_0(t)y(t) = g(t)$$

This ODE is of order n if $f_n(t) \neq 0$.

Reduction of order

Solve $\mathsf{T}y = 0$ for the case where $c = \dfrac{b^2}{4}$

- Method of trial solution only gives one solution, $y(t) = \mathrm{e}^{-\frac{b}{2}t}$
- Trial solution of the form, $y(t) = u(t)\mathrm{e}^{-\frac{b}{2}t}$
- $y' = \left(u' - \dfrac{b}{2}u\right)\mathrm{e}^{-\frac{b}{2}}$ and $y'' = \left(u'' - bu' + \dfrac{b^2}{4}u\right)\mathrm{e}^{-\frac{b}{2}}$
- $\mathsf{T}y = \left(u'' - bu' + \dfrac{b^2}{4}u\right)\mathrm{e}^{-\frac{b}{2}} + b\left(u' - \dfrac{b}{2}u\right)\mathrm{e}^{-\frac{b}{2}} + \dfrac{b^2}{4}u\mathrm{e}^{-\frac{b}{2}} = 0$
- $u'' = 0$, so $u = C_1t + C_2$ and hence we have found the general solution

$$y(t) = (C_1t + C_2)\,\mathrm{e}^{-\frac{b}{2}t}$$

1. Given known complementary solution, $v(t)$, to a second order ODE, write general solution as $y(t) = u(t)v(t)$
2. This reduces the order of the ODE to one: $w' + \left(\dfrac{2v'}{v} + f_1\right)w = \dfrac{g}{v}$ where $w = u'$
3. Use the integrating factor $I = \mathrm{e}^{\int 2\frac{v'}{v} + f_1 \mathrm{d}t}$ to solve for w.
4. This leads to the general solution $y = v\int\left[\dfrac{1}{I}\int\dfrac{Ig}{v}\mathrm{d}t\right]\mathrm{d}t$

Finding the first solution for $\mathsf{T}v = 0$ is usually the difficult task.

Theorem

An IVP for a 2nd order linear ODE, $(\mathsf{T}y)(t) = g(t)$, with $\mathsf{T} = \frac{\mathrm{d}^2}{\mathrm{d}t^2} + f_1(t)\frac{\mathrm{d}}{\mathrm{d}t} + f_0(t)$, and specified initial values $y(t_0)$ and $y'(t_0)$ has a unique solution $y(t)$, which is twice differentiable for all t on any interval which contains t_0 and over which f_0, f_1 and g are continuous.

Linear algebra: two vectors, $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$, are linearly dependent if there exists c such that $\boldsymbol{v}_1 = c\boldsymbol{v}_2$.

Differential equations: two functions, y_1 and y_2, are linearly dependent if there exists c such that $y_1 = cy_2$ for all t.

Example

$y_1 = \mathrm{e}^{-2t}$ and $y_2 = 2\mathrm{e}^{-2t}$ are linearly dependent because $y_1 = \frac{1}{2}y_2$

Definition (The Wronskian)

Given two functions y_1 and y_2, the Wronskian is given by

$$\mathrm{wr}[y_1, y_2](t) = \det\begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix} = y_1y_2' - y_1'y_2$$

- If there exists a t such that $\mathrm{wr}[y_1, y_2](t) \neq 0$ then y_1 and y_2 are linearly independent.
- If y_1 and y_2 are linearly dependent then $\mathrm{wr}[y_1, y_2](t) = 0$ for all t.
- $\mathrm{wr}[y_1, y_2] = 0$ for all t does not imply y_1 and y_2 are linearly dependent.

Aside: If the two functions are in the kernel of $\mathsf{T} = \frac{\mathrm{d}^2}{\mathrm{d}t^2} + f_1\frac{\mathrm{d}}{\mathrm{d}t} + f_0$, i.e., $\mathsf{T}y_1 = 0$ and $\mathsf{T}y_2 = 0$, then $\mathrm{wr}[y_1, y_2](t) \propto \mathrm{e}^{-\int f_1 \mathrm{d}t}$, i.e., if the Wronskian is zero anywhere, it is zero everywhere.

Exercise 1.6

In two-dimensional electrostatics, we encounter the following ODE (which is also known as the Euler equation):

$$\frac{\mathrm{d}^2\phi}{\mathrm{d}r^2} + \frac{1}{r}\frac{\mathrm{d}\phi}{\mathrm{d}r} - \frac{n^2}{r^2}\phi = 0, \qquad r > 0, \qquad n \in \{0, 1, 2, \ldots\}^*$$

Solve this ODE by trialling functions of the form $\phi = r^\lambda$.

General form $(\mathsf{T}y)(t) = g(t)$ where

$$\mathsf{T} = \frac{\mathrm{d}^n}{\mathrm{d}t^n} + f_{n-1}(t)\frac{\mathrm{d}^{n-1}}{\mathrm{d}t^{n-1}} + \ldots + f_1(t)\frac{\mathrm{d}}{\mathrm{d}t} + f_0(t).$$

Our linear algebra analogy works again.

Theorem (Solutions of an nth order linear ODE/IVP)

- ODEs: The general solution to $\mathsf{T}y = g$ is the sum of n linearly independent complementary solutions and a single particular solution.
  - complementary solutions form an n-dimensional vector space.
- IVPs: If the above ODE is supplemented with initial values $y(t_0), y'(t_0), \ldots, y^{(n-1)}(t_0)$ it has a unique solution which is n-times differentiable for all t on any interval in which $f_0, f_1, \ldots, f_{n-1}$ and g are continuous.

Definition (The Wronskian)

Given n functions $y_1, y_2, \ldots, y_n$, the Wronskian is given by

$$\mathrm{wr}[y_1, y_2, \ldots, y_n](t) = \det\begin{pmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{pmatrix}$$

- If there exists a t such that $\mathrm{wr}[y_1, y_2, \ldots, y_n](t) \neq 0$ then $y_1, y_2, \ldots, y_n$ are linearly independent.
- If any of $y_1, y_2, \ldots, y_n$ are linearly dependent then $\mathrm{wr}[y_1, y_2, \ldots, y_n](t) = 0$ for all t.
- $\mathrm{wr}[y_1, y_2, \ldots, y_n](t) = 0$ for all t for all t does not imply any of $y_1, y_2, \ldots, y_n$ are linearly dependent.

1. An nth order ODE has n *linearly independent* complementary solutions and a single particular solution (the kernel of the linear transformation has dimension n)
2. If initial values are specified, $y(t_0), y'(t_0), \ldots, y^{n-1}(t_0)$ then this IVP has a unique solution (subject to continuity constraints): slide 30
3. Check for linear independence using the Wronskian: slide 31
4. If one complementary solution for a 2nd order ODE is known, the general solution can be found through reduction of order: slide 20

When does it fall apart?

- If you can't find a complementary solution to your ODE.

Let $\mathbf{A}$ be a diagonalisable matrix (recall your linear algebra),

$$\mathbf{A} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}$$

where $\mathbf{D}$ is a diagonal matrix whose entries are the eigenvalues and the columns of $\mathbf{S}$ contain the eigenvectors.
Then

$$\begin{aligned} \boldsymbol{y}' + \mathbf{S}\mathbf{D}\mathbf{S}^{-1}\boldsymbol{y} &= \boldsymbol{b} \\ \mathbf{S}^{-1}\boldsymbol{y}' + \mathbf{D}\mathbf{S}^{-1}\boldsymbol{y} &= \mathbf{S}^{-1}\boldsymbol{b} \\ \boldsymbol{z}' + \mathbf{D}\boldsymbol{z} &= \mathbf{S}^{-1}\boldsymbol{b} \end{aligned}$$

where $\boldsymbol{z} = \mathbf{S}^{-1}\boldsymbol{y}$. This system is decoupled and can now be solved as n independent first-order linear ODEs.

1. Higher order ODEs can be written as a single *vector-valued* ODE.
2. Constant coefficient homogeneous: use exponential trial functions.
3. Constant coefficient inhomogeneous: use matrix diagonalisation to decouple the system then use an integrating factor on individual ODEs.

When does it fall apart?

- Matrix is not diagonalisable or coefficients are functions of t: try matrix exponentials.

Definition (Taylor series)

- Let $f(t)$ be N-times differentiable. Then

$$P_N(t) = \sum_{n=0}^{N} \frac{f^{(n)}(a)}{n!}(t-a)^n$$

is the Nth order Taylor polynomial.

- If $f(t)$ is infinitely differentiable, then the Taylor series of f is

$$P(t) = \lim_{N\to\infty} P_N(t) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(t-a)^n$$

Theorem (Convergence of a power series)

For the power series $\sum_{n=0}^{\infty} c_n(t-a)^n$, we define

$$l = \lim_{n\to\infty}\left|\frac{c_{n+1}}{c_n}\right|.$$

l^{-1} is the *radius of convergence*, i.e.,

- The power series converges for $|t-a| < l^{-1}$,
- The power series diverges for $|t-a| > l^{-1}$.
- If $l = 0$ then the power series converges everywhere and if $l = \infty$ the power series diverges everywhere except $t = a$.

If we require the convergence at $|t-a| = l^{-1}$, we must check using another test.

1. Substitute $y(t) = \sum_{n=0}^{\infty} c_n t^n$ into ODE.
2. Change variable in each sum to have the same power of t.
3. Extract any low index terms so all sums are over the same range.
4. Solve for the coefficients using recurrence relations.
5. Check if you have the right number of linearly independent solutions.
   - We differentiated our power series term by term to find the derivative of a series
   - Is this allowed? I.e.,

$$\frac{\mathrm{d}}{\mathrm{d}t}\sum_{n=0}^{\infty} c_n t^n \overset{?}{=} \sum_{n=0}^{\infty}\frac{\mathrm{d}}{\mathrm{d}t}c_n t^n$$

Definition (Uniform convergence)

The sequence of functions $W_N(t)$ converges *uniformly* to the function $W(t)$ on domain D if

$$\lim_{N\to\infty}\max_{t\in D}|W_N(t) - W(t)| = 0,$$

i.e., the *maximum* error in the Nth approximation goes to zero as $N \to \infty$.

Definition (Uniform convergence of a series)

The series $\sum_{n=1}^{\infty} w_n(t)$ converges uniformly to $W(t)$ on domain D if the sequence $W_N(t) = \sum_{n=1}^{N} w_n(t)$ converges uniformly to $W(t)$ on D, i.e.,

$$\lim_{N\to\infty}\max_{t\in D}\left|\sum_{n=1}^{N} w_n(t) - W(t)\right| = 0$$

Theorem (Weierstrass test for uniform convergence)

Let $w_n(t), n = 1, 2, 3, \ldots$ be functions on domain D. If $\max_{t\in D}|w_n(t)| \leqslant a_n$ for some $a_n \in \mathbb{R}$ such that $\sum_{n=1}^{\infty} a_n$ converges then $\sum_{n=1}^{\infty} w_n(t)$ converges *uniformly* on D.

Theorem (Integrating a series)

Consider a sequence of *continuous* functions $w_n(t), n = 0, 1, 2, \ldots$ defined on a closed interval $D = [a, b]$. If $\sum_{n=0}^{\infty} w_n(t)$ converges *uniformly* on D, then

1. $$\int_{t_0}^{t}\sum_{n=0}^{\infty} w_n(\tau)\mathrm{d}\tau = \sum_{n=0}^{\infty}\int_{t_0}^{t} w_n(\tau)\mathrm{d}\tau, \qquad t_0, t \in D$$
   i.e., the series can be integrated term by term
2. The integrated series is also uniformly convergent on D.

Theorem (Differentiating a series)

Consider a sequence of continuously differentiable functions $w_n(t), n = 0, 1, 2, \ldots$ defined on a closed interval $D = [a, b]$. If

1. the series $\sum_{n=0}^{\infty}\frac{\mathrm{d}}{\mathrm{d}t}w_n(t)$ converges uniformly on D and
2. the series $\sum_{n=0}^{\infty} w_n(t)$ converges pointwise for some $t_0 \in D$

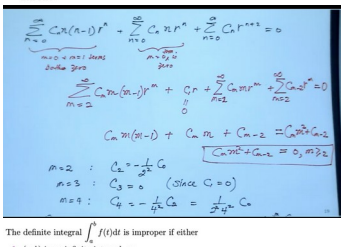
then the series $\sum_{n=0}^{\infty} w_n(t)$ satisfies the following properties;

1. it converges uniformly on D,
2. it is differentiable on D and
3. it can be differentiated term by term by $\frac{\mathrm{d}}{\mathrm{d}t}\sum_{n=0}^{\infty} w_n(t) = \sum_{n=0}^{\infty}\frac{\mathrm{d}}{\mathrm{d}t}w_n(t)$.

- We can search for a *formal* power series solution to an ODE

$$y(t) = \sum_{n=0}^{\infty} c_n t^n$$

- This power series converges uniformly for any domain strictly within its radius of convergence
- This power series can be differentiated term by term
- The power series solution is a *genuine solution* if the radius of convergence is non-zero.



The definite integral $\int_a^b f(t)\mathrm{d}t$ is improper if either

1. (a, b) is an infinite interval, or
2. f is not bounded on (a, b).

Assume all unboundedness occurs at a and/or b (otherwise split up the integral). The improper integral is then defined as

$$\int_a^b f(t)\mathrm{d}t = \lim_{r\to a^+}\lim_{s\to b^-}\int_r^s f(t)\mathrm{d}t$$

Definition (Comparison test)

Let f and g be two continuous functions on (a, b).

1. If $|f(t)| \leqslant g(t)$ for all $t \in (a, b)$ and $\int_a^b g(t)\mathrm{d}t$ converges then $\int_a^b f(t)\mathrm{d}t$ converges.
2. If $f(t) \geqslant g(t) \geqslant 0$ for all $t \in (a, b)$ and $\int_a^b g(t)\mathrm{d}t$ diverges then $\int_a^b f(t)\mathrm{d}t$ diverges.

The Laplace transform

The *Laplace transform*, $\mathscr{L}$, is a linear transformation that maps a suitable function, $f(t)$, defined on $(0, \infty)$ to a new function $(\mathscr{L}f)(s)$:

$$(\mathscr{L}f)(s) = \int_0^\infty \mathrm{e}^{-st} f(t)\mathrm{d}t.$$

Since it is a *linear transformation*:

1. $(\mathscr{L}\{f + g\})(s) = (\mathscr{L}f)(s) + (\mathscr{L}g)(s)$
2. $(\mathscr{L}\{\alpha f\}) = \alpha(\mathscr{L}f)(s)$ where α is a constant

$\mathscr{L}f$ involves an improper integral, which doesn't always converge.

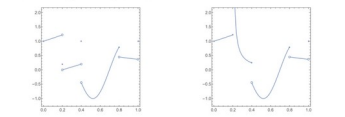
A set of criteria to ensure that $\mathscr{L}f$ *converges* is

- f is piecewise-continuous on $[0, L]$ for all $L > 0$.
- $|f(t)| \leqslant ke^{\alpha t}$ for all $t > T$ where there exist $\alpha, k, T \in \mathbb{R}$, i.e., f cannot grow faster than some exponential function.

Piecewise continuous?

- f has a finite number of discontinuities on $(0, L)$.
- The limits of f at the end points, and on each side of every discontinuity, are finite.

Are the following functions piecewise continuous?



Theorem

Let F be the vector space of all *continuous* functions on $[0, \infty)$ which have a Laplace transform. Then,

$$(\mathscr{L}f)(s) = 0 \quad \text{for all} \quad s > \alpha \quad \Rightarrow \quad f(t) = 0 \quad \text{for all} \quad t \geqslant 0,$$

i.e., the inverse transform exists, $(\mathscr{L}^{-1}(\mathscr{L}f))(t) = f(t)$ for $t \in [0, \infty)$.

Lessons from linear algebra:

1. If the kernel of a linear transformation has dimension zero then the linear transformation is invertible.
2. If a linear transformation is invertible then the inverse transformation is also a linear transformation

*A formula for the inverse exists but it requires complex analysis from MAST30021

1. If f is continuous at t,
$$(\mathscr{L}^{-1}(\mathscr{L}f))(t) = f(t)$$
2. If f is discontinuous at $t = 0$,
$$(\mathscr{L}^{-1}(\mathscr{L}f))(0) = \lim_{a\to 0^+} f(a)$$
3. If f is discontinuous at $t, t > 0$,
$$(\mathscr{L}^{-1}(\mathscr{L}f))(t) = \frac{1}{2}\left(\lim_{a\to t^-} f(a) + \lim_{a\to t^+} f(a)\right)$$
i.e., the inverse Laplace transform returns the midpoint at the discontinuity.

$f(t)$	$(\mathscr{L}f)(s)$	
1	$\dfrac{1}{s}$,	$s > 0$
$\mathrm{e}^{\alpha t}$	$\dfrac{1}{s-\alpha}$	$s > \alpha$
$\sin\alpha t$	$\dfrac{\alpha}{s^2+\alpha^2}$	$s > 0$
$\cos\alpha t$	$\dfrac{s}{s^2+\alpha^2}$	$s > 0$

$f(t)$	$(\mathscr{L}f)(s)$	
$\sinh\alpha t$	$\dfrac{\alpha}{s^2-\alpha^2}$	$s > \alpha$
$\cosh\alpha t$	$\dfrac{s}{s^2-\alpha^2}$	$s > \alpha$
$t^n,\ n \in \mathbb{N}$	$\dfrac{n!}{s^{n+1}}$	$s > 0$

Theorem (Laplace transform of the derivative)

Let f be a continuous function on $[0, \infty)$. If $\mathscr{L}f$ exists then

$$(\mathscr{L}f')(s) = s(\mathscr{L}f)(s) - f(0).$$

The initial condition is encoded into the Laplace transform!

Proof:

Theorem (Higher order derivatives)

Let $f, f', \ldots, f^{(n-1)}$ be continuous on $[0, \infty)$ and have Laplace transforms. Then

$$(\mathscr{L}f^{(n)})(s) = s^n(\mathscr{L}f)(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \ldots - f^{(n-1)}(0)$$

Idea of proof:

Can be proved by induction using integration by parts.

Example

We will use the case of a second derivative extensively:

$$(\mathscr{L}f'')(s) = s^2(\mathscr{L}f)(s) - sf(0) - f'(0)$$

Theorem (s-shift)

If $\mathscr{L}f$ exists for $s > \alpha$, then

$$(\mathscr{L}\mathrm{e}^{\alpha t}f)(s) = (\mathscr{L}f)(s - \alpha) \quad s > \alpha + \omega$$

Definition (Heaviside step function)

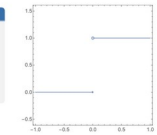
$$H(t) = \begin{cases} 0 & t \leqslant 0 \\ 1 & t > 0 \end{cases}$$

The value of $H(0)$ is usually irrelevant.

Its translation is defined

$$H_c(t) = H(t - c), \quad c \geqslant 0$$

It can be used to model turning processes on or off.



Theorem (t-shift)

If $\mathscr{L}f$ exists for $s > \alpha$, and $c \geqslant 0$, then

$$(\mathscr{L}f(t-c)H_c(t))(s) = \mathrm{e}^{-cs}(\mathscr{L}f)(s), \quad s > \alpha$$

Exercise 4.16

Find the Laplace transform of $f = H_3(t)\sin(t-3)$

The Dirac delta

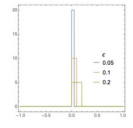
Consider the piecewise function

$$\delta_\epsilon^+ = \frac{H_0(t) - H_\epsilon(t)}{\epsilon} = \begin{cases} 0 & t \leqslant 0 \\ \frac{1}{\epsilon} & 0 < t \leqslant \epsilon, \\ 0 & t > \epsilon \end{cases} \qquad \epsilon > 0$$

It is useful for modelling when something noticeable happens over a negligible time period, i.e., an impulse.

The 'strength' of this impulse is

$$\int_0^\infty \delta_\epsilon^+\,\mathrm{d}t = \int_0^\epsilon \frac{1}{\epsilon}\,\mathrm{d}t = 1$$



1. $\lim_{\epsilon\to 0}\delta_\epsilon^+ = 0$ for all $t \neq 0$.
2. $\lim_{\epsilon\to 0}\int_0^\infty \delta_\epsilon^+\,\mathrm{d}t = 1$.
3. $\lim_{\epsilon\to 0}(\mathscr{L}\delta_\epsilon^+) = \lim_{\epsilon\to 0}\dfrac{1 - \mathrm{e}^{-\epsilon s}}{\epsilon s} = 1$ (L'Hôpital's rule)

No function can have these properties!

Definition

We define $\delta^+(t)$ such that

1. $\delta^+(t) = 0$ for all $t \neq 0$,
2. $\displaystyle\int_0^\infty \delta^+\mathrm{d}t = 1$ and
3. $\mathscr{L}\delta^+ = 1$.

While δ^+ is not a function*, it is extremely useful.

- Since δ^+ is zero for $t \neq 0$, it "samples" the value of a continuous function at $t = 0$
$$f(t)\delta^+(t) = f(0)\delta^+(t), \quad \text{where } f(t) \text{ is continuous at } t = 0$$
- If $f(t)$ is piecewise continuous, then
$$f(t)\delta^+(t) = f(0^+)\delta^+(t), \quad \text{where } f(0^+) = \lim_{t\to 0^+} f(t).^*$$
- The most useful form of this relation is
$$\int_0^\infty f(t)\delta^+(t)\mathrm{d}t = f(0^+)$$
- Define $\delta_c^+(t) = \delta^+(t - c)$ for $c \geqslant 0$, then
  1. $\displaystyle\int_0^\infty f(t)\delta_c^+(t)\mathrm{d}t = f(c^+)$ where $f(c^+) = \lim_{t\to c^+} f(t)$,
  2. $(\mathscr{L}\delta_c^+)(s) = \displaystyle\int_0^\infty \mathrm{e}^{-st}\delta_c^+(t)\mathrm{d}t = \mathrm{e}^{-sc}$.

Definition (Convolution)

Given two functions, f and g, defined on $[0, \infty)$, their convolution, $(f * g)(t)$ is defined

$$(f * g)(t) = \int_0^t f(t-\tau)g(\tau)\mathrm{d}\tau^\dagger$$

†subject to this integral existing

- Convolutions are commutative, i.e., $(f * g)(t) = (g * f)(t)$
- Convolutions are not like multiplication, $1 * f \neq f$
$$(1 * f)(t) = \int_0^t f(\tau)\mathrm{d}\tau$$
i.e., convolving with 1 is equivalent to antidifferentation.

Theorem (Laplace transform of the convolution)

Let f and g be functions with Laplace transforms. Then $f * g$ exists and has Laplace transform

$$(\mathscr{L}f * g) = (\mathscr{L}f)(\mathscr{L}g)$$

i.e., Laplace transforms convert convolutions into multiplication.

We can use the previous two theorems to calculate the Laplace transform of an antiderivative by convoluting the function with 1.

Lemma (Laplace transform of an antiderivative)

$$\left(\mathscr{L}\int_0^t f(\tau)\mathrm{d}\tau\right) = (\mathscr{L}1 * f) = (\mathscr{L}1)(\mathscr{L}f) = \frac{\mathscr{L}f}{s}$$

Recall that for a symmetric (or hermitian or self-adjoint) matrix, $\mathbf{A} \in \mathbb{R}^{n\times n}$, using the dot product as the inner product, i.e., $\langle \boldsymbol{u}, \boldsymbol{v}\rangle = \boldsymbol{u}\cdot\boldsymbol{v}$, then the following hold:

1. The eigenvalues, λ, and eigenvectors, $\boldsymbol{v}$, of $\mathbf{A}$ satisfy $\mathbf{A}\boldsymbol{v} = \lambda\boldsymbol{v}$.
2. The eigenvalues, λ, of $\mathbf{A}$ are real,
3. The eigenvectors, $\boldsymbol{v}$, of $\mathbf{A}$ can be chosen to be orthonormal,
4. These n orthonormal eigenvectors, $\boldsymbol{v}_i$, are a basis of $\mathbb{R}^n$.
5. Any vector, $\boldsymbol{u} \in \mathbb{R}^n$, can be written as a unique linear combination of the eigenvectors
$$\boldsymbol{u} = \sum_{i=1}^{n} c_i\boldsymbol{v}_i, \qquad \text{where} \qquad c_i = \langle \boldsymbol{u}, \boldsymbol{v}_i\rangle = \boldsymbol{u}\cdot\boldsymbol{v}_i$$

Recall that the solutions to $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + ay = 0$ are $y = C_1\sin\sqrt{a}x + C_2\cos\sqrt{a}x$ for $a > 0$.

- $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + ay = 0$ defines an eigenvalue problem for the linear operator $-\frac{\mathrm{d}^2}{\mathrm{d}x^2}$, where a are the *eigenvalues* and y are the *eigenfunctions*.*
- If we want to represent periodic functions, we need some additional conditions
$$y(-L) = y(L) \qquad \text{and} \qquad y'(-L) = y'(L)$$
These are called *periodic boundary conditions*. We will cover other boundary conditions in more detail in topic 5.

Inner products

In exercise 4.1, we found the following eigenvalues and eigenfunctions

for $\lambda_0 = 0$, $y_0 = \dfrac{1}{\sqrt{2}}$

for $\lambda_n = k_n^2 = \dfrac{n^2\pi^2}{L^2}$, $y_n^{(1)} = \cos k_n x$ and $y_n^{(2)} = \sin k_n x$ where $n = 1, 2, 3, \ldots$.

If we want to use our linear algebra framework, we need an inner product. Define

$$\langle f, g\rangle = \frac{1}{L}\int_{-L}^{L} fg\,\mathrm{d}x$$

Example

The inner product of x with x^3 is $\langle x, x^3\rangle = \frac{1}{L}\int_{-L}^{L} x^4\mathrm{d}x = \frac{1}{L}\left[\frac{x^5}{5}\right]_{-L}^{L} = \frac{2L^4}{5}$.

On slide 6, how did we know $\lambda \in \mathbb{R}$? Recall self-adjoint from slide 4

Theorem (Self-adjoint linear transformations)

Let T be a linear transformation. T is self-adjoint if

$$\langle \mathsf{T}f, g\rangle = \langle f, \mathsf{T}g\rangle.$$

- The eigenvalues of T are real.
- $\mathsf{T} = -\frac{\mathrm{d}^2}{\mathrm{d}x^2}$ with periodic boundary conditions is self adjoint under the inner product on Slide 8.

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Any function, $s(x)$, in the vector-space can be written as a linear combination (*Fourier series*) of the eigenfunctions

$$s(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos k_n x + b_n \sin k_n x \qquad \text{where} \qquad k_n = \frac{n\pi}{L}.$$

- Each term in the Fourier series is called a *mode*.
- a_n and b_n are the *amplitude* of the mode.
- k_n is the mode's *wave number*.
- If independent variable is time we use angular frequency, ω_n, not wave number.

- The *Fourier series* of a function, $f(x)$, which is defined over $(-L, L)$, is given by

$$s(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos k_n x + b_n \sin k_n x \qquad \text{where} \qquad k_n = \frac{n\pi}{L}$$

- The *Fourier coefficients* are given by

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos k_n x \,\mathrm{d}x \qquad (\text{substitute } k_0 = 0 \text{ to calculate } a_0)$$
$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin k_n x \,\mathrm{d}x$$

- We are NOT saying $s(x) = f(x)$

Theorem (Pointwise convergence of a Fourier series)

Let f be a piecewise continuous function defined over $[-L, L]$, whose derivative, f', is also piecewise continuous (f is known as piecewise smooth). Then:

- The Fourier series of f exists, i.e., the integrals defining the Fourier coefficients all converge.
- The Fourier series converges to the value, $F(x)$, of the periodic extension of $f(x)$ for every x where $F(x)$ is continuous.
- For all x where $F(x)$ is discontinuous, the Fourier series instead converges to

$$\frac{1}{2}\left(\lim_{w\to x^-} F(w) + \lim_{w\to x^+} F(w)\right)$$

$$\lim_{N\to\infty} \max_{t\in D} \left|\sum_{n=1}^{N} w_n(t) - W(t)\right| = 0$$

Theorem (Weierstrass test for uniform convergence)

Let $w_j(t)$, $j = 1, 2, 3, \ldots$ be functions on domain D. If $\max_{t\in D}|w_j(t)| \leqslant a_j$ for some $a_j \in \mathbb{R}$ such that $\sum_{j=1}^{\infty} a_j$ converges then $\sum_{j=1}^{\infty} w_j(t)$ converges *uniformly* on D.

- A uniformly convergent series can be integrated term by term
- No guarantee a uniformly convergent series can be differentiated term by term

$$f_1(x) = x^2 \qquad \Rightarrow \qquad s_1(x) = \frac{\pi}{3} + 4\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2}\cos nx$$

$$f_2(x) = x \qquad \Rightarrow \qquad s_2(x) = -2\sum_{n=1}^{\infty}\frac{(-1)^n}{n}\sin nx$$

- For x^2 the coefficients scale as $\frac{1}{n^2}$ but for x, they scale as $\frac{1}{n}$.
- The function with the continuous periodic extension converges faster than the discontinuous one.

Theorem

If f' is piecewise-continuous on $-L < x \leqslant L$, and the periodic extension of f is continuous on $\mathbb{R}$, then the Fourier series of f exists and it converges *uniformly* to the periodic extension of f on $\mathbb{R}$.

- If the periodic extension of f is discontinuous, but otherwise satisfies the conditions for pointwise convergence of a Fourier series (slide 19) then the convergence will NOT be uniform.

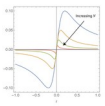
Definition (Gibbs phenomenon)

The Fourier series of a discontinuous function overshoots and rings on either side of the discontinuity with a maximum error of approximately 8.95% of the gap height.

- Recall the plots on Slide 22. As we increase the number of terms, it appears the error between the Fourier series and the periodic function near $x = \pm\pi$ does not approach zero.
- The Fourier series of x exhibits Gibbs phenomena.

1. In Exercise 4.8 differentiating term by term gave the correct pointwise convergence, but there is no guarantee this will happen.
2. Consider the sequence of functions

$$W_N(x) = \frac{x}{1 + N^2x^2}.$$



$W_N(x) \to W(x)$ uniformly over $[-L, L]$ for $L > 0$ where $W(x) = 0$.

3. $W'(x) = 0$ and

$$W_N'(x) = \frac{1 - N^2x^2}{(1 + N^2x^2)^2}$$

but $\lim_{N\to\infty} W_N(x) \to \begin{cases} 1, & x = 0 \\ 0, & x \neq 0 \end{cases}$.

1. Let $f(x)$ be even, i.e., $f(-x) = f(x)$. Then all b_n terms are zero and the Fourier series of $f(x)$ is

$$s(x) = \frac{1}{2}a_0 + \sum_{n=0}^{\infty} a_n \cos k_n x \quad \text{where} \quad a_n = \frac{2}{L}\int_0^L f(x)\cos k_n x \mathrm{d}x$$

2. Let $f(x)$ be odd, i.e., $f(-x) = -f(x)$. Then all a_n terms are zero and the Fourier series of $f(x)$ is

$$s(x) = \sum_{n=1}^{\infty} b_n \sin k_n x \quad \text{where} \quad b_n = \frac{2}{L}\int_0^L f(x)\sin k_n x \mathrm{d}x$$

3. For a function defined on $[0, L]$ we can extend it over $[-L, L]$ by making it either odd or even and then using 1 and 2 above.

Remember $k_n = \frac{n\pi}{L}$

Instead of using sines and cosines, we could use complex exponentials. This leads to the complex Fourier series

$$s(x) = \sum_{n=-\infty}^{n=\infty} c_n e^{in\pi x/L}, \qquad c_n = \frac{1}{2L}\int_{-L}^{L} f(x)e^{-in\pi x/L}\mathrm{d}x.$$

This will be our starting point for Fourier transforms later in semester.

Initial value problem: conditions applied at a single point

Boundary value problem (BVP): conditions applied at multiple points

- Conditions can be on;
  1. the solution, e.g., $y(2) = 2$,
  2. the derivatives, e.g., $y'(4) = 0$,
  3. a mix of both, e.g., $y(1) + 2y'(1) - y''(1) = 2$,
  4. a mix of different points, e.g., $y(2) + y(3) = 0$.
- Boundary conditions may also be non-linear, e.g., $y^2(2)y'(2) + \sin y(2) = 0$.

Definition (Linear boundary value problem)

A 2nd order linear boundary value problem consists of a 2nd order linear ODE

$$f_2(x)y''(x) + f_1(x)y'(x) + f_0(x) = g(x)$$

and two linear boundary conditions;

$$a_1y'(x_1) + b_1y(x_1) = c_1 \qquad \text{and} \qquad a_2y'(x_2) + b_2y(x_2) = c_2.$$

- If $c_1 = c_2 = 0$ then the boundary conditions are homogeneous.
- If $c_1 = c_2 = 0$ and $g(x) = 0$ then the boundary value problem is homogeneous.

- Dirichlet boundary conditions fix the value, e.g.,

$$y(0) = 0 \quad \text{and} \quad y(L) = 1.$$

- Neumann boundary conditions fix the derivative, e.g.,

$$y'(0) = 0 \quad \text{and} \quad y'(L) = 0.$$

- Mixed boundary conditions contain both Dirichlet and Neumann conditions, e.g.,

$$y'(-L) = 0 \quad \text{and} \quad y(0) + y'(0) = 1$$

Unlike linear IVPs, linear BVPs are not guaranteed a unique solution (or even that a solution exists).

1. The complementary solutions to a homogeneous BVP form a vector space.
   - This vector space may be zero dimensional (i.e., zero is the only solution, see case 1 from exercise 5.1).
2. The solutions to a general linear boundary value problem are obtained by adding a single particular solution to the complementary solutions.
3. If no particular solution exists, then the general linear BVP has no solutions.

The solutions to BVPs play a key role in [illegible] partial differential equations.

1. The *order* of a PDE is the order of the highest derivative.
2. A PDE is *linear* if it is a linear combination of partial derivatives of the unknown function, whose coefficients may only be functions of the independent variables, which equals a function of the independent variables. If it does not meet this criteria it is *non-linear*.
3. A PDE is *homogeneous* if each term contains the unknown function. It is *inhomogeneous* if there is a term which contains only the independent variables.

- General solutions of ODEs have unknown constants, which we try to specify with initial or boundary conditions.
- General solutions of PDEs can contain *unknown functions*.
- If the PDE is first order in time, we usually constrain the unknown function to be some specified function, $u(\boldsymbol{x}, t)$ at the initial time, i.e., $u(\boldsymbol{x}, 0) = f(\boldsymbol{x})$.
- If it is second order in time, we impose an additional constraint on the *time* derivative of the unknown function, i.e., $\partial_t u(\boldsymbol{x}, 0) = g(\boldsymbol{x})$.
- Boundary conditions are constraints imposed at specified locations in space for all times after the initial condition.
- Let D be a bounded spatial domain, then boundary conditions are imposed on the domain's boundary, which is denoted ∂D.
- Two most common boundary conditions are:
  1. *Dirichlet boundary conditions* fix the value of the unknown function on the boundary, i.e., $u(\boldsymbol{x}, t) = f(\boldsymbol{x}, t)$ where $\boldsymbol{x} \in \partial D$.
  2. *Neumann boundary conditions* fix the value of the normal derivative at the boundary, i.e., $\partial_{\boldsymbol{n}} u(\boldsymbol{x}, t) = g(\boldsymbol{x}, t)$ where $\boldsymbol{x} \in \partial D$ and $\boldsymbol{n}$ is the boundary's outwards unit normal.

If a PDE has appropriate initial and boundary conditions, it is called an *initial-boundary value problem* (IBVP).

Solving a PDE using separation of variables is a lengthy process involving many steps:

1. Separate variables to turn single PDE into multiple ODEs.
2. Apply boundary conditions to find eigenvalues/eigenfunctions.
3. Solve remaining ODEs for each eigenvalue.
4. Apply Superposition Principle to get a series general solution.
5. Apply initial conditions to find the series coefficients (leads to a Fourier series or similar).

We modify the previous problem so that the whole road (including the ends) is insulated. This is modelled through *Neumann* boundary conditions: $\partial_x u(0, t) = \partial_x u(L, t) = 0$.

We apply the initial condition $u(x, 0) = f(x) = 1 - \cos k_1 x$ where $k_1 = \frac{\pi}{L}$.



1. Separate variables to turn single PDE into multiple ODEs.
   - The exact procedure as our previous example leads to

$$-X''(x) = \lambda X(x) \qquad -T'(t) = \lambda\alpha T(t)$$

2. Apply boundary conditions to find eigenvalues/eigenfunctions
   - Substituting seperable solution into the boundary conditions gives

$$\partial_x u(0, t) = X'(0)T(t) = 0 \qquad \text{and} \qquad \partial_x u(L, t) = X'(L)T(t) = 0$$

   and hence $X'(0) = X'(L) = 0$.
   - To solve our eigenvalue problem we again consider three cases, because our linear transformation is self-adjoint under these boundary conditions and inner product $\langle f, g\rangle = \frac{2}{L}\int_0^L f\,g\,\mathrm{d}x$.
   - $\lambda = 0$ gives $X(x) = C_1x + C_2$. Both boundary conditions require $C_1 = 0$ while C_2 is arbitrary. Hence $\lambda_0 = 0$ is an eigenvalue with eigenfunction $X_0(x) = 1$.
   - $\lambda < 0$ gives no non-trivial solutions.
   - $\lambda = k^2 > 0$ gives eigenvalues $\lambda_n = k_n^2$ where $k_n = \frac{n\pi}{L}$ and eigenfunctions $X_n(x) = \cos k_n x$
3. Solve remaining ODEs for each eigenvalue
   - $\lambda_n = k_n^2$ gives the same solutions as previous, i.e., $T_n(t) = \mathrm{e}^{-\alpha k_n^2 t}$ and $\lambda_0 = 0$ gives

$$T_0'(t) = -\alpha\lambda_0 T(t) = 0 \qquad \Rightarrow \qquad T_0(t) = a_0$$

4. Superposition to get series general solution.
   - We sum all our solutions to get

$$u(x, t) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \mathrm{e}^{-\alpha k_n^2 t}\cos k_n x$$

5. Apply initial conditions to fix series coefficients.

$$u(x, 0) = 1 - \cos\frac{\pi x}{L} = 1 - \cos k_1 x = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos k_n x$$

- This time we have a Fourier cos series (this is why we put the 1/2 factor in front of the constant term). We use orthogonality by inspection to see

$$a_0 = 2, \qquad a_1 = -1 \qquad \text{and} \qquad a_n = 0 \quad \text{for} \quad n \neq 1$$

- The final solution is therefore

$$u(x, t) = 1 - \mathrm{e}^{-\alpha k_1 t}\cos k_1 x$$

- A solution which is constant in time is called a *steady-state* solution, u_s.
- A solution which decays to zero for $t \to \infty$ is called a *transient* solution, u_t.
- On Slide 32, $u_s = 1$ and $u_t = -\mathrm{e}^{-\alpha k_1 t}\cos k_1 x$
- We can find steady-state solutions by trialling a solution which is independent of t and satisfies the boundary conditions.

Separation of variables usually won't work with inhomogeneities: two types to consider

1. Inhomogeneous boundary conditions, e.g., $\partial_x u(0, t) = \sin(t)$
2. Inhomogeneous PDEs , e.g., $\dfrac{\partial u}{\partial t} = \alpha\dfrac{\partial^2 u}{\partial x^2} + \mathrm{e}^{-t}$

In both cases we follow the approach from ODEs:

Split solution into a part which satisfies the homogeneous problem, and a particular solution which satisfies the relevant inhomogeneity.

- The particular solution satisfies

$$\frac{\partial}{\partial t}u_s(x) = \alpha\frac{\partial^2}{\partial x^2}u_s(x) \qquad \Rightarrow \qquad \frac{\mathrm{d}^2 u_s}{\mathrm{d}x^2} = 0$$

- The solution is a linear function and the boundary conditions give

$$u_s(x) = x$$

- The homogeneous solution now satisfies

$$\frac{\partial}{\partial t}(u_t + u_s) = \alpha\frac{\partial^2}{\partial x^2}(u_t + u_s) \qquad \Rightarrow \qquad \frac{\partial u_t}{\partial t} = \alpha\frac{\partial^2 u_t}{\partial x^2}$$

- The boundary conditions are

$$u(0, t) = u_t(0, t) + u_s(0) = 0 \qquad \Rightarrow \qquad u_t(0, t) = 0$$
$$u(\pi, t) = u_t(L, t) + u_s(L) = L \qquad \Rightarrow \qquad u_t(L, t) = 0$$

- The homogeneous solution is a *transient* solution.
- The initial condition is

$$u(x, 0) = u_t(x, 0) + u_s(x)$$
$$0 = u_t(x, 0) + x \qquad \Rightarrow \qquad u_t(x, 0) = -x$$

- The homogeneous problem can now be solved by separation of variables.
- Dirichlet boundary conditions lead to the homogeneous general solution

$$u_t(x, t) = \sum_{n=1}^{\infty} b_n \mathrm{e}^{-\alpha k_n^2 t}\sin k_n x$$

- Applying the initial condition of the homogeneous problem gives

$$u_t(x, 0) = -x = \sum_{n=1}^{\infty} b_n \sin k_n x$$

- We calculated the Fourier sin series of x in Topic 4, so the coefficients for $-x$ are:

$$b_n = \frac{2(-1)^n L}{n\pi}$$

- The complete solution is therefore

$$u(x, t) = u_s(x) + u_t(x, t) = x + \frac{2L}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^n}{n}\mathrm{e}^{-\alpha k_n^2 t}\sin k_n x$$

1. Separate variables to turn single PDE into multiple ODEs.
   - $u(x, y) = X(x)Y(y)$

$$X''(x)Y(y) + X(x)Y''(y) = 0 \qquad \Rightarrow \qquad \frac{X''(x)}{X(x)} = -\frac{Y''(y)}{Y(y)} = -\lambda$$
$$-X''(x) = \lambda X(x) \qquad \text{and} \qquad Y''(y) = \lambda Y(y)$$

2. Apply boundary conditions to find eigenvalues/eigenfunctions.
   - Not always obvious which boundary conditions to start with. In x we have homogeneous Neumann conditions and in y inhomogeneous Dirichlet conditions.
   - Start with the homogeneous ones

$$\partial_x u(0, y) = X'(0)Y(y) = 0 \qquad \text{and} \qquad \partial_x u(1, y) = X'(1)Y(y) = 0$$
$$\Rightarrow \qquad X'(0) = X'(1) = 0$$

   The eigenvalues/eigenfunctions are $\lambda_0 = 0$, $X_0(x) = 1/\sqrt{2}$ and

$$\lambda_n = n^2\pi^2, \quad X_n(x) = \cos n\pi x, \quad \text{for} \quad n = 1, 2, 3, \ldots \quad \text{and} \quad \langle f, g\rangle = 2\int_0^1 f\,g\,\mathrm{d}x$$

3. Solve remaining ODEs for each eigenvalue.
   - $Y_0(y) = \frac{1}{2}(Ay + B)$ and

$$Y_n''(y) = n^2\pi^2 Y_n(y) \quad \text{gives} \quad Y_n(y) = C_n^{(1)}\cosh n\pi y + C_n^{(2)}\sinh n\pi y$$

4. Superposition to get series general solution.
   - The general solution is therefore:

$$u(x, y) = \frac{1}{2}(Ay + B) + \sum_{n=1}^{\infty}\left(C_n^{(1)}\cosh n\pi y + C_n^{(2)}\sinh n\pi y\right)\cos n\pi x$$

5. Apply ~~initial~~ *additional boundary* conditions to fix series coefficients.
   - The homogeneous Dirichlet condition gives

$$u(x, 0) = \frac{1}{2}B + \sum_{n=1}^{\infty} C_n^{(1)}\cos n\pi x = 0 \qquad \Rightarrow \qquad B = 0 \quad \text{and} \quad C_n^{(1)} = 0$$

   - The remaining boundary conditions gives

$$u(x, 1) = x^2(3 - 2x) = \frac{1}{2}A + \sum_{n=1}^{\infty} C_n^{(2)}\sinh n\pi\cos n\pi x$$

   Let $a_n = C_n^{(2)}\sinh n\pi$ then we have a regular Fourier cos series. First, $A = 2\int_0^1 x^2(3 - 2x)\mathrm{d}x = 1$

$$a_n = 2\int_0^1 x^2(3 - 2x)\cos n\pi x \,\mathrm{d}x \qquad \Rightarrow \qquad a_n = -\frac{24(1 - (-1)^n}{n^2\pi^2}$$
$$a_{2m-1} = -\frac{48}{(2m-1)^4\pi^4} \qquad \Rightarrow \qquad C_{2m-1}^{(2)} = -\frac{48}{(2m-1)^4\pi^4\sinh n\pi}$$

The complete solution is therefore

$$u(x, y) = y - \sum_{m=1}^{\infty}\frac{48}{(2m-1)^4\pi^4}\frac{\sinh(2m-1)\pi y}{\sinh(2m-1)\pi}\cos(2m-1)\pi x$$

To show a solution to a PDE is a genuine solution, i.e., that it satisfies the PDE, we have to show we can bring all derivatives inside the infinite sum.

1. Choose a domain to demonstrate uniform convergence, e.g., $x \in [0, L]$ and $t > T$ where $L, T \in \mathbb{R}_{>0}$)
2. Show that one of the terms in the PDE converges uniformly (the other term will then also converge uniformly)
3. Show the solution converges at some point in the domain
4. Show all derivatives up to the ones in the PDE converge at some point in the domain, i.e., for a PDE with $\frac{\partial^2 u_n}{\partial t^2}$ terms show $\frac{\partial u_n}{\partial t}$ converges at a point.

- Consider the solution to the inhomogeneous heat equation on Slide 42. Let's twice differentiate each term wrt x:

$$\frac{\partial^2 u_n}{\partial x^2} = -\frac{2(-1)^n(L^2 - n^2\pi^2\alpha)}{L\alpha n\pi}\mathrm{e}^{-k_n^2\alpha t}\sin k_n x + \frac{2(-1)^n\pi^2}{L^2\alpha n}\sin k_n x,$$

which does not converge uniformly due to the $\frac{1}{n}$ decay in the steady state term.

- Formal solutions which don't satisfy the PDE are an example of *weak solutions*.
- Weak solutions are extremely useful in both numerical and analytical analysis of differential equations. You will meet them again in advanced differential equations courses.
- After proving a solution satisfies the PDE (genuine solution) we also need to consider if it is unique, i.e., the only solution to the IBVP.
- All PDEs covered in MAST20030 produce unique solutions.
- The proof is not examinable but the general idea is to show that given two solutions, u_1 and u_2, then $v = u_1 - u_2$ must be zero everywhere. This is usually done through an *energy functional*: $E[v](t) = \int_D v(x, t)^2\mathrm{d}x$ where D is the spatial domain. If $E[v](t) = 0$ for all t then the solution is unique.

Instead of using sines and cosines, we could use complex exponentials. This leads to the complex Fourier series

$$s(x) = \sum_{n=-\infty}^{n=\infty} c_n e^{in\pi x/L}, \qquad c_n = \frac{1}{2L}\int_{-L}^{L} f(x)\mathrm{e}^{-in\pi x/L}\mathrm{d}x.$$

This is our starting point for Fourier transforms later in semester.

Takes a suitable function, $f(x)$ defined on $x \in \mathbb{R}$ and maps it to a new function, $(\mathscr{F}f)(k)$, usually defined for $k \in \mathbb{R}$.

$$(\mathscr{F}f)(k) = \widehat{f}(k) = \int_{-\infty}^{\infty} f(x)\mathrm{e}^{-ikx}\mathrm{d}x \qquad \text{and} \qquad f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\widehat{f}(k)\mathrm{e}^{ikx}\mathrm{d}k$$

- The Fourier transform is a linear transformation.
  1. $(\mathscr{F}\{f + g\})(k) = (\mathscr{F}f)(k) + (\mathscr{F}g)(k)$
  2. $(\mathscr{F}\{\alpha f\})(k) = \alpha(\mathscr{F}f)(k)$

Theorem (Existence and inversion of the Fourier transform)

Let $f : \mathbb{R} \to \mathbb{C}$ be a piecewise continuous function with finitely many critical points on any closed interval within its domain such that $\int_{-\infty}^{\infty}|f(x)|\mathrm{d}x$ converges.

Then the Fourier transform of $f(x)$ exists and,

1. if $f(x)$ is continuous at x, then
$$\left(\mathscr{F}^{-1}\widehat{f}\right)(x) = f(x)$$
2. if $f(x)$ is discontinuous at x, then
$$\left(\mathscr{F}^{-1}\widehat{f}\right)(x) = \frac{1}{2}\left(\lim_{\xi\to x^-} f(\xi) + \lim_{\xi\to x^+} f(\xi)\right)$$

$f(x)$	$\widehat{f}(k)$
$\delta_a(x),\ a \in \mathbb{R}$	e^{-iak}
$\mathrm{e}^{-a\lvert x\rvert},\ a > 0$	$\dfrac{2a}{a^2 + k^2}$
$H_{-a}(x) - H_a(x),\ a > 0$	$2a\,\mathrm{sinc}\,ak$
$\mathrm{e}^{-ax^2},\ a > 0$	$\sqrt{\dfrac{\pi}{a}}\mathrm{e}^{-\frac{k^2}{4a}}$

$f(x)$	$\widehat{f}(k)$
$\mathrm{e}^{ibx},\ a \in \mathbb{R}$	$2\pi\delta_b(x)$
$\dfrac{1}{a^2 + x^2},\ a > 0$	$\dfrac{\pi}{a}\mathrm{e}^{-a\lvert k\rvert}$
$\mathrm{sinc}\,ax,\ a > 0$	$\dfrac{\pi}{a}(H_{-a}(x) - H_a(x))$
$\mathrm{sech}\,ax,\ a \neq 0$	$\dfrac{\pi}{a}\mathrm{sech}\,\dfrac{\pi k}{2a}$

*$\mathrm{sinc}(x) = \dfrac{\sin x}{x}$ for $x \neq 0$ and $\mathrm{sinc}(0) = 1$

Theorem (The Fourier transform of a convolution)

We define the convolution of $f(x)$ with $g(x)$ as

$$(f * g)(x) = \int_{-\infty}^{\infty} f(x - x')g(x')\mathrm{d}x'$$

Then the Fourier transform of the convolution is

$$(\mathscr{F}\{f * g\})(k) = (\mathscr{F}f)(k)(\mathscr{F}g)(k)$$

Note this is different to the definition we used in Laplace transforms because we are working on $x \in \mathbb{R}$ instead of $t > 0$.

1. $(\mathscr{F}\{f(x - y)\})(k) = \mathrm{e}^{-iky}(\mathscr{F}f)(k)$, for $y \in \mathbb{R}$.
2. $(\mathscr{F}\{\mathrm{e}^{iqx}f\})(k) = (\mathscr{F}f)(k - q)$, for $q \in \mathbb{R}$.

Examples

$$\left(\mathscr{F}\mathrm{e}^{-a|x-2|}\right)(k) = \frac{2a\mathrm{e}^{-2ik}}{a^2 + k^2}$$

$$\left(\mathscr{F}^{-1}\mathrm{sech}\frac{\pi(k-5)}{2}\right)(x) = \mathrm{e}^{5ix}\mathrm{sech}\,x$$

We've already seen that the Fourier transform of $f'(x)$ is $ik\widehat{f}(k)$.

1. Differentiation w.r.t. x turns into multiplication by k
$$\left(\mathscr{F}\frac{\mathrm{d}^n f}{\mathrm{d}x^n}\right)(k) = (ik)^n(\mathscr{F}f)(k)$$
2. Multiplication by x turns into differentiation w.r.t. k
$$(\mathscr{F}x^n f)(k) = i^n\frac{\mathrm{d}^n}{\mathrm{d}k^n}(\mathscr{F}f)(k)$$

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